

MANUEL RODRÍGUEZ VILLEGAS

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PROFILE

AI engineer with international research and industry experience and a strong mathematical background. My approach starts with first-principles reasoning to turn complex problems into simple, efficient solutions. I bring energy and enjoy fast-paced environments where I can take ownership and meaningful responsibility.

EDUCATION

Comillas Pontifical University, ETSI ICAI <i>Master's Degree in Artificial Intelligence; GPA: 9.08/10; Class Representative</i> – Ongoing Master's thesis: Vision-only autonomous drones with differentiable physics.	Madrid, Spain 2025 – 2026
Cornell University <i>Exchange Student - Electrical & Computer Engineering</i>	Ithaca, NY, USA Jan – May 2025
Comillas Pontifical University, ETSI ICAI <i>BE in Mathematical Engineering and AI; GPA: 8.75/10</i>	Madrid, Spain 2021 – 2025

EXPERIENCE

AUDI AG <i>Geometric AI Intern</i> – Developing 3D defect classification under limited data and strong confounding. – Building AI agent skills for NLP-based requirements classification and conflict detection, and reliable, low-latency, and token-efficient access to a graph database with 100,000+ records.	Ingolstadt, Germany July 2026 – Nov 2026
Comillas Pontifical University <i>Teaching Assistant</i> – Contributed to deployment of a GPU-based AI cluster to support student workloads in deep learning courses.	Madrid, Spain Sep 2025 – Apr 2026
Azzulei Technologies <i>Computer Vision Intern</i> – Built a real-time multi-object tracking pipeline (YOLOv8 + Optical Flow) for automated sports camera control, achieving robust performance at 30 FPS.	Madrid, Spain Jun – Aug 2025
Imperial College London <i>Research Assistant</i> – Reproduced nonlinear dynamical systems using Neural ODEs and PINNs (PyTorch), validating stability and trajectory accuracy against ground-truth simulations and different initial conditions.	Remote (London, UK) Jun – Aug 2024
Endesa <i>Energy Data Analyst Intern</i> – Forecasted daily Iberian ancillary service costs using MLP and regression models (MAPE < 10%).	Madrid, Spain Jun – Aug 2023

SELECTED RESEARCH

TopoSIGMA: Spectral Information-Geometric Mapping Architecture – Designed an online, training-free ROS 2 mapping system separating spectral place discovery from covariance-aware loop closure, using DINOv2 descriptors and directional node distributions. – Manuscript in preparation for submission to Robotics and Autonomous Systems.	2026
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SKILLS

Software: Python (advanced), PyTorch, JAX, ROS 2, Git, Docker, K8S, LangGraph, n8n
AI & ML: Deep Learning, Generative & Agentic AI (MCP), Geometric AI, Computer Vision, NLP, MLOps, DRL
Mathematics: Linear Algebra, Probabilistic AI, Optimization, Differential Geometry, Statistics

LANGUAGES

Spanish (Native) | English (Fluent) | German (Beginner)

HONORS & AWARDS

Bending Spoons Hackathon Winner <i>Selected among Spain's top 20 tech students for First Ascent and winner of the event's hackathon.</i>	Milan, Italy 2026
Hackathon Winner <i>Developed the winning project in the 10th Smart Industry Hackathon, featuring Kearney and CAF.</i>	Madrid, Spain 2025
UNIJES Social Tech Challenge Winner <i>Voice-controlled wheelchair project</i>	Bilbao, Spain 2024
Academic Excellence Scholarship <i>Top-scoring students across Community of Madrid</i>	Madrid, Spain 2021, 2023, 2024
Baccalaureate Honorable Mention <i>2nd highest GPA in Community of Madrid</i>	Madrid, Spain 2021